

Tejaswini Peddireddy

AI/ML Engineer | Computer Science Undergraduate

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SUMMARY

AI/ML Engineer and Computer Science undergraduate (GPA 9.35/10.00) specializing in Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing (NLP). Experienced in building and deploying end-to-end machine learning solutions, including CNN-based image classification models (98% test accuracy), YOLOv8 object detection pipelines, and NLP-based text classification systems. Proficient in Python, TensorFlow, Keras, PyTorch, and Scikit-learn, with hands-on experience in data preprocessing, model development, hyperparameter tuning, and model deployment on edge devices (Raspberry Pi) and cloud platforms (AWS). Seeking an AI/ML Engineer or Machine Learning Engineer role to apply deep learning, computer vision, and NLP expertise to real-world production systems.

EDUCATION

B.Tech in Computer Science and Engineering (AI & ML) — Presidency University

Bengaluru, Karnataka | 2023–2027 (Expected) | GPA: 9.35 / 10.00

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, HTML, CSS
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, PyTorch, CNN, Transfer Learning, ResNet50, MobileNetV2, Model Evaluation, Hyperparameter Tuning
- **Reinforcement Learning:** Q-Learning, SARSA, Monte Carlo (MC) Control, TD Learning
- **Computer Vision:** OpenCV, YOLOv8, Image Processing, Image Classification, Object Detection
- **NLP:** Text Preprocessing, Tokenization, NLTK, TF-IDF, Text Classification, Confusion Matrix
- **Data & Visualization:** NumPy, Pandas, Data Analysis, Matplotlib
- **Tools & Platforms:** Git, GitHub, Google Colab, Anaconda, Visual Studio Code, Raspberry Pi, Roboflow
- **Cloud:** AWS
- **Core Concepts:** Object-Oriented Programming (OOP), Model Deployment, MLOps Fundamentals

WORK EXPERIENCE

Web Development Intern — Fitbiz Solutions

June 2025 – July 2025

- Developed and maintained 3 key pages of a live production website (fitnessfactory.fit) using HTML, CSS, and WordPress, ensuring cross-device compatibility across desktop, tablet, and mobile.
- Tested and refined responsive UI components across 3 screen breakpoints (desktop, tablet, mobile), improving website usability and reducing layout inconsistencies.

PROJECTS

Bell Pepper Leaf Disease Detection using Transfer Learning

[GitHub ↗](#)

Python, TensorFlow, Keras, ResNet50, MobileNetV2, OpenCV

- Developed a CNN-based leaf disease classification model using a dataset of 8,500+ labeled images.
- Implemented data preprocessing, augmentation, and transfer learning using pre-trained CNN architectures (ResNet50, MobileNetV2).
- Achieved 98% test accuracy while significantly reducing false positives, enabling scalable automated plant disease detection.

AI-Powered Precision Spraying Drone for Smart Agriculture

[GitHub ↗](#)

Python, YOLOv8, OpenCV, Flask, Raspberry Pi, Picamera2, GPIO, Roboflow

- Deployed an AI-powered precision spraying system using YOLOv8, OpenCV, and Raspberry Pi for real-time crop monitoring and leaf disease detection.

- Fine-tuned a custom YOLOv8 object detection model on an annotated leaf disease dataset using Roboflow and Google Colab for real-time inference.
- Designed a Flask web dashboard to stream live camera feeds, visualize detections, and automate pesticide spraying via GPIO control.

AI-Powered Multilingual Video Dubbing System

[GitHub ↗](#)

Python, Raspberry Pi, FFmpeg, Speech Recognition, Google Translator API, gTTS

- Engineered a Raspberry Pi-based multilingual video translation system converting English videos into Indian regional languages with dubbed audio and synchronized subtitles.
- Integrated speech recognition, machine translation, and text-to-speech (TTS) APIs to automate end-to-end multilingual video localization.
- Supported multiple Indian regional languages, improving accessibility of video content for diverse audiences.

Friends Dialogue Classification using NLP

[GitHub ↗](#)

Python, Scikit-learn, NLTK, Pandas, TF-IDF

- Built an NLP text classification model to identify speakers from Friends TV series dialogue using a full preprocessing pipeline (tokenization, stopword removal, lemmatization, TF-IDF).
- Evaluated and compared multiple ML models, assessing performance using accuracy, precision, recall, and F1-score.

CERTIFICATIONS

- [Introduction to Large Language Models — NPTEL](#)
- [Mathematical Foundations of Machine Learning — NPTEL](#)
- [Social Networks — NPTEL](#)
- [Building with the Claude API — Anthropic](#)
- [Python Course for Beginners — Scalar](#)

ACHIEVEMENTS

- Secured a Top 70 rank among 502 teams in a Raspberry Pi project competition.
- Participated in Chip to Crop Hackathon — Presidency University.
- Participated in L&T Hackathon and Smart India Hackathon (SIH).
- Served as anchor for multiple college-level sports events.
- Competed in the All India University (AIU) South Zone Yoga Championship.